

PROJECT SUTRA — PRECISION DISASTER PITCH SCRIPT

REAL-WORLD DISASTER DATA (WAYANAD 2024, KEDARNATH 2024) • HARIKA • NIKHIL

PRESENTERS: HARIKA • NIKHIL
TOTAL TIME: ~2 MIN 15 SEC

< LATEST 2024 FIELD DATA: Cites actual after-action reports from the July 2024 Wayanad Landslides (Chooralmala/Mundakkai) and the August 2024 Kedarnath Valley cloudbursts.

SLIDE 01 Introduction: Project SUTRA

00:00 - 00:30 (30s)

SPEAKER: HARIKA (OPERATIONS LEAD) [STAGE: CENTER STAGE • COMPELLING OPENING HOOK]

HOOK: "When over 300 lives were buried under mud in the July 2024 Wayanad landslides, and 15,000 pilgrims were trapped in Kedarnath gorges this August, rescue teams faced a critical 72-hour survival window. But the moment standard drones entered thick forests and mountain ravines, signals collapsed and drones went blind."

"Respected jury, we are **Team OFFGRID**. I am Harika, and with me is Nikhil, our Tech Lead. We built **Project SUTRA** — an autonomous drone swarm engineered to find survivors in disaster zones where **GPS fails and communications collapse**."

"Instead of single manual drones, SUTRA deploys an autonomous swarm backed by 3 breakthroughs:"

PILLAR 1 Decentralized Flight: Drones dodge obstacles and coordinate together at 50Hz without ground towers. [Slide: 50Hz GNC]

PILLAR 2 Machine-Learning Comms: Live video stays connected through deep mountain ravines without blacking out. [Slide: -5dB JSCC]

PILLAR 3 Thermal & 3D AI Vision: Fuses body heat sensors with 3D terrain mapping to spot survivors through smoke and pinpoint their exact coordinates on steep slopes. [Slide: Subsys C • 3D Vision]

CONTINUES WITH HARIKA → "Let me show you why today's search and rescue missions fail in the field."

SLIDE 02 The Problem: 4 Real Disaster Bottlenecks

00:30 - 01:15 (45s)

SPEAKER: HARIKA [STAGE: URGENT, FACTUAL DISASTER AUDIT]

HOOK: "When our team audited actual NDRF and Army after-action reports from 2024 disaster zones, we identified 4 fatal bottlenecks:"

VOID 1 (GNC) Canopies Block GPS: In the **July 2024 Wayanad Landslides** (Mundakkai & Chooralmala), dense plantation canopies blocked satellite lock. Commercial drones drifted into trees, suffering a **70% canopy crash rate**.

VOID 2 (COMMS) Mountain Ridges Cut Video: In the **August 2024 Kedarnath Valley washouts**, 3,000-meter mountain ridges severed radio line-of-sight. Standard digital video crashed below 5dB SNR into instant blackouts.

VOID 3 (VISION) Flat-Earth Errors on Steep Slopes: In 45-degree slope washouts (like the **2024 Shimla & Sikkim floods**), standard 2D drone cameras assumed flat ground, miscalculating survivor coordinates by **15 to 30 meters**.

VOID 4 (C2 GCS) Human Pilot Bottleneck: Deploying 5 separate drones required **15 to 25 crew members** and 2 hours of setup — burning ₹12.5 Lakhs per mission while survivors ran out of air.

CONTINUES WITH HARIKA → "Here is how SUTRA's 4 core moats solve each breakdown."

SPEAKER: HARIKA [STAGE: HIGH-ENERGY SOLUTION PRESENTATION]

HOOK: "SUTRA solves each void through 4 hardened engineering moats:"

- MOAT 1Self-Flying Swarms: Drones calculate collision-free flight paths onboard at 50Hz — zero ground towers, zero GPS reliance.
[Slide: SUTRA-FSD & 3D ORCA]
- MOAT 2ML-Enabled Video Link: When signals weaken behind mountain ridges, our neural encoder transmits clear video down to -5dB SNR without digital blackouts. [Slide: Deep JSCC | -5dB]
- MOAT 3Thermal & 3D AI Vision: Fuses thermal body heat with 3D elevation raycasting on Jetson Orin (<15ms), giving rescue teams exact survivor coordinates on steep slopes. [Slide: 3D DEM • 3.59cm Residual]
- MOAT 4One-Person Tactical Command: An offline 3D tactical map lets 1 operator command the entire swarm for just ₹95,000 per sortie. [Slide: Pegasus 3D GCS]

"The bottom line: SUTRA scans a full square mile in 10 to 18 minutes — 3 to 4 times faster than traditional teams, saving lives when every second counts."

Metric	Traditional SAR	Single Drone	SUTRA SWARM
Area Coverage / mi ²	2-3 hours	45-60 min	10-18 min (4x Faster)
Detection Rate	65-70%	55-65%	78-85% First Pass
Personnel Required	15-25 people	2-3 pilots	1-2 Operators
Sortie Cost	₹12.5 Lakhs	₹5.0 Lakhs	₹95,000

🔥 POWER HANDOFF TO NIKHIL → "Now, our Tech Lead, Nikhil, will walk you through the deep engineering behind each subsystem and our real-world field deployment."

STAGE TAKEOVER: NIKHIL (TECH LEAD) [DEEP SUBSYSTEM DEEP DIVES, CODE ARCHITECTURE & FIELD CONOPS]

- Slide 04 (Subsystem A): 50Hz PX4 MicroXRCE Offboard + 3D ORCA Velocity Obstacles
- Slide 05 (Subsystem B): Deep JSCC Autoencoders & Mesh Resiliency at -5dB SNR
- Slide 06 (Subsystem C): Tri-Modal Cross-Attention (RGB + FLIR Thermal + Radar) + WGS84 3D DEM Geolocation
- Slide 07 (Subsystem D): WebGPU 3D Digital Twin & Offline ATAK Field Command
- Slide 08 (Field CONOPS & Q&A): Wayanad Simulation Audit, Hardware BOM, and Jury Defense